

ANGEL DARLA

Health Informatics & AI/ML Data Analyst | Machine Learning · Deep Learning · Healthcare Analytics

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Health Informatics graduate student (CGPA: 9.17/10) with a strong technical foundation in Python, machine learning, and deep learning, building predictive models, NLP pipelines, and healthcare data systems. Currently co-developing an MSc research project on clinically-guided feature selection (DODA) for multi-disease risk prediction. Maintains an active portfolio of 15+ applied AI/ML projects spanning computer vision, LLM fine-tuning, FastAPI deployment, and healthcare interoperability standards (FHIR/ABDM). Clinical training sharpens problem framing and domain validation in health-tech applications, with a growing research interest in bias and fairness in medical AI.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, Exploratory Data Analysis, Data Cleaning

Machine Learning: Scikit-learn, Predictive Modeling, Logistic Regression, Feature Selection, Model Evaluation, SHAP Explainability

Deep Learning: PyTorch, U-Net (Image Segmentation), CNNs, Transfer Learning, Active Learning

NLP & LLMs: spaCy, NLTK, Hugging Face Transformers, LoRA/PEFT Fine-Tuning, Prompt Engineering, Ollama

Visualization & BI: Power BI, Streamlit, Gradio, Matplotlib

Backend & Deployment: FastAPI, REST API Development, SQLite, Git/GitHub

Healthcare Data Standards: FHIR, ABDM (Ayushman Bharat Digital Mission), EHR, Clinical Decision Support Systems (CDSS)

Tools: Jupyter, Google Colab, Hugging Face Hub

EDUCATION

M.Sc. Health Informatics & Analytics — The Apollo University, Chittoor 2025 – Present

CGPA: 9.17 / 10

Coursework: Clinical Decision Support Systems, Healthcare Data Standards & Interoperability, Health Data Analytics, Applied Machine Learning

B.Sc. Nursing — Dr. N.T.R. University of Health Sciences 2021 – 2025

FEATURED PROJECTS

DODA: Clinically-Guided Feature Selection for Multi-Disease Risk Prediction | Ongoing MSc Research · Python, Scikit-learn, SHAP — [GitHub](#)

- Co-developing (with a research partner) a post-hoc, selector-independent clinical knowledge layer that re-weights statistically selected features across four diseases (breast cancer, CKD, diabetes, heart disease) to recover clinically important predictors missed by conventional feature selection.
- Built a modular pipeline benchmarking six selectors (ANOVA, LASSO, mRMR, Random Forest Importance, Boruta, Mutual Information) under repeated stratified cross-validation, with stability metrics (Jaccard similarity, rank correlation) and SHAP-based interpretability comparison.

ABDM CareBridge — Continuity-of-Care Framework for Migrant Health Records | Python, FHIR, Probabilistic Record Linkage — [GitHub](#)

- Built a reference implementation of an ABDM-enabled interoperability framework supporting continuity of care for interstate migrant workers, using synthetic FHIR resources.
- Simulated end-to-end patient healthcare journeys and applied probabilistic record linkage to match identities across disconnected facility records.

Heart Disease Risk Prediction API | Python, Machine Learning, FastAPI — [GitHub](#)

- Built and deployed a machine learning model to predict heart disease risk from patient clinical data.
- Developed a FastAPI service to serve real-time predictions, demonstrating end-to-end model deployment.

Retinal Vessel Segmentation | Python, PyTorch, U-Net — [GitHub](#)

- Implemented a baseline U-Net model for retinal vessel segmentation on fundus images as a hands-on medical imaging deep learning project.

Llama 3.2 LoRA Fine-Tuning | *Python, Hugging Face, PEFT, LoRA* — [GitHub](#)

- Fine-tuned Meta Llama 3.2 (1B) using parameter-efficient LoRA adapters, with training, evaluation, and base-vs-adapter performance comparison.

Clinical Notes NLP Pipeline | *Python, spaCy, NLTK* — [GitHub](#)

- Built an NLP pipeline to process unstructured clinical notes, extracting entities, symptoms, and medications.
- Applied POS tagging and visualization techniques to surface structured healthcare insights.

Breast Cancer Risk Prediction | *Python, Logistic Regression, Scikit-learn* — [GitHub](#)

- Developed a reproducible logistic regression pipeline with stratified train-test splitting and feature standardization.
- Evaluated performance using confusion matrix and classification metrics.

ADDITIONAL PROJECTS

- Fingerprint Feature Analyzer — AI-powered fingerprint image feature extraction and visualization using computer vision, deep learning, and FastAPI. (GitHub: fingerprint-feature-analyzer)
- Clinical Readmission Analytics — Statistical analysis and EDA on a 25,000-record diabetic readmission dataset to uncover clinical insights. (GitHub: clinical-readmission-analytics)
- Hospital ER Dashboard — Interactive Power BI dashboard visualizing ER patient flow, wait times, admissions, and department workload. (GitHub: Hospital-ER-Dashboard)
- Healthcare ETL Pipeline — End-to-end ETL pipeline structuring raw patient records into a SQLite database for analytics. (GitHub: healthcare-etl-pipeline)
- Chest X-Ray AI Annotation — AI-assisted lung segmentation and uncertainty-based annotation using U-Net and active learning. (GitHub: chest-xray-ai-annotation)

RESEARCH INTERESTS

AI in Healthcare • Medical Data Analysis • Bias & Fairness in Medical AI • Clinical Feature Selection & Explainability

CERTIFICATIONS

- AI in Medical Diagnosis — Coursera
- Python for Data Analysis — Analytics Vidhya
- Transforming Healthcare with IT — Conference, Hyderabad

STRENGTHS

- Analytical Thinking — Interprets healthcare datasets and translates them into clinically meaningful insights.
- Applied ML Execution — Consistently ships end-to-end projects: data → model → deployed API/dashboard.
- Interdisciplinary Perspective — Clinical training combined with health informatics and data science.
- High Learning Initiative — Rapidly expanding into deep learning, NLP, and LLM fine-tuning through self-directed projects.

REFERENCES

Dr. R. Satyanarayana — Head of Department, School of Health Sciences, The Apollo University, Chittoor
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